Summary of the Two Most Recent Trials

TASK-026 (tuning and data-scaling diagnostic) and TASK-027 (sample-efficiency ideas)
Aim: gate G1.2 of Phase 1 on Craftax-Classic

Branch `claude/t026-sample-efficiency`; trials of 2026-10-19 and 2026-10-22

Prepared 2026-10-23 for review

How to read this document

Thank you for taking the time to review these results. The document is organised around figures; each caption states what to look at. **Colour code used throughout:** ■ P5-clean arm (counts as EP-A evidence), ■ P5-contested arm (reward coefficient changed; *cannot* count toward any gate), ■ diagnostic only (not EP-A). **Metrics** are `reward_pct` (mean of distinct achievements/22 × 100) and `score_pct` (Crafter score). **Gate G1.2** requires $mean - 2SE > 55.49$ (reward) and > 16.77 (score) with 10 seeds, 999,424 env steps, 1345-d symbolic observation (EP-A). Section 1 gives the whole story on two pages; later sections give the evidence; the appendices hold every number.

1 At a glance

Bottom line. Neither trial reached G1.2; Phase 1 remains open and Phase 2 remains locked. The gap is on one metric: score already passes, reward does not. TASK-026 localised the gap (it is *sample efficiency*, not capacity); TASK-027 tried to close it and moved the clean arm by only +0.3 reward, while the one lever that helped is reward shaping, which the protocol (P5) does not allow to count.

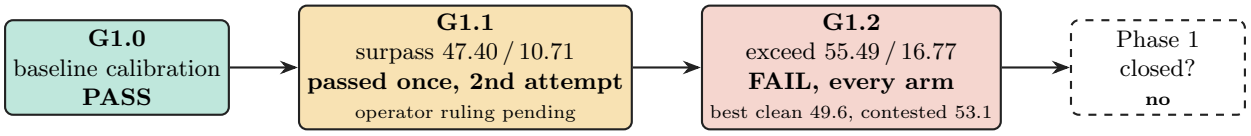


Figure 1: Gate status after both trials. A phase closes only when *every* gate passes under EP-A (roadmap rule `phase_advance`). G1.S (EP-B screening) was not run.

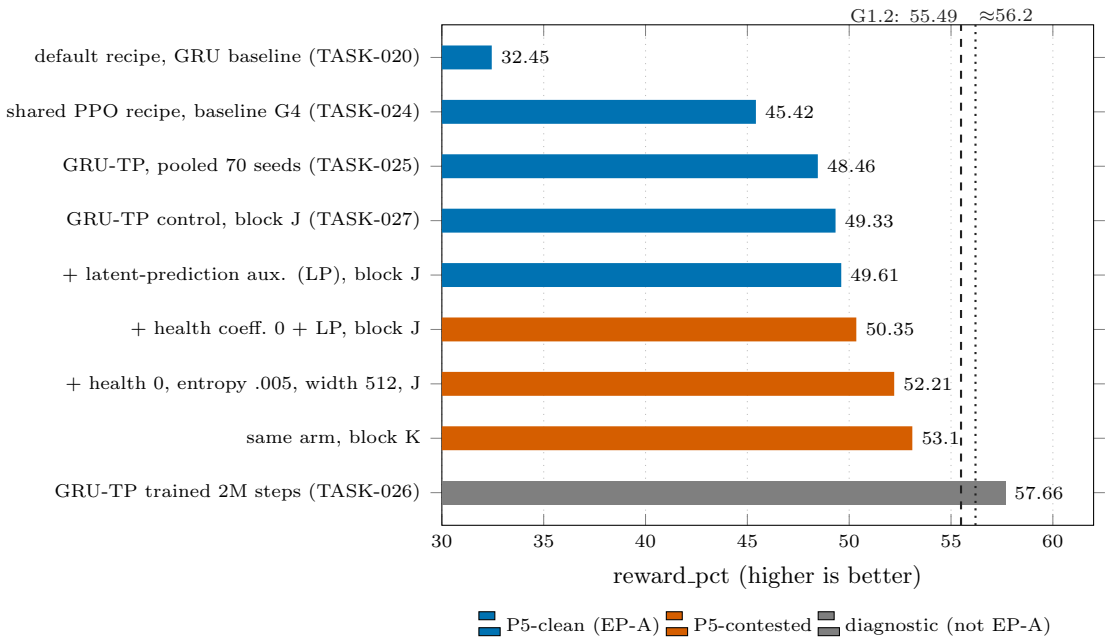


Figure 2: How far each arm is from the G1.2 line. Dashed: the paper value 55.49. Dotted: the mean a low-variance arm needs so that mean − 2 SE exceeds 55.49 (≈ 56.2, quoted from the TASK-026 report). The 2M-step bar is a data-scaling diagnostic (2 seeds, different step budget), shown only to indicate where the gap closes. Values are the means reported for each row; spreads are in Section 4.

The two trials side by side.

	TASK-026 (2026-10-19)	**TASK-027** (2026-10-22)
Question	Can ten single-knob changes of GRU-TP close the 7.7 reward gap? Where does the gap come from?	Can an lr schedule, a learned reward coefficient, or parameter changes raise sample efficiency toward G1.2?
Platform	cloud session, 4 CPU cores, jax 0.12.0	Kaggle T4, jax 0.11.1 (screens and finals)
Seeds	tuning 4000–4003 (R0), 4000–4001 (variants)	tuning 4000–4003, 5000–5003, 6000–6003; finals 172–181 (J), 182–191 (K)
Final EP-A block?	none (pre-registered stop line 54.0 not reached)	yes: four arms in block J, one arm also in block K
Outcome	no knob qualifies; **2M steps crosses G1.2** ⇒ gap is sample efficiency	clean arm $+0.28 \pm 0.60$ reward; contested best 52.21 / 53.10; **G1.2 fails**

2 Common setting and protocols

Label	**Used for**	**Rules (abridged)**
`EP-A`	gate evidence	999,424 env steps, 10 seeds (all listed), first episode of 256 fresh envs per seed, sampled policy, final checkpoint, clean commit, tuning on disjoint seeds
`EP-A-tuning`	screening	same budget and metrics, 2–4 seeds, tuning seeds only; never a gate
`diag-*`	diagnostics	other step budgets (2M/3M/4M); never a gate, never quoted beside paper numbers
`EP-A-P5-contested-reward-coefficient`	reward-coefficient arms	EP-A mechanics, but the *training* reward is altered, which violates prohibition P5; MLflow `gate_eligible=false`

Reference arm	**GRU-TP** (registry `Base_002_D04_S00`, config `G4TP1`): GRU baseline with LayerNorm and skip heads (G4), plus the transition-prediction (TP) auxiliary head. 1.254M params total, 0.880M deployed. It is the control in both trials.
Thresholds (G1.2)	MFRL_best, Dedieu et al. 2025 Table 1: 55.49 reward, 16.77 score (55.6M params, image observation; modality caveat recorded).
Starting point	Best clean arm before these trials: GRU-TP pooled over 70 seeds 48.46±0.15 reward, 18.51 ± 0.21 score. Score already clears 16.77; reward needs about +7.7 (about 1.6 more achievements per episode).

3 TASK-026: tuning attempt and data-scaling diagnostic

3.1 Ten single-knob variants: none helps

Pre-registered rule: a variant *qualifies* if it beats R0 by +2.0 reward on the same seeds, and a final EP-A block is run only if some variant exceeds a 54.0 stop line.

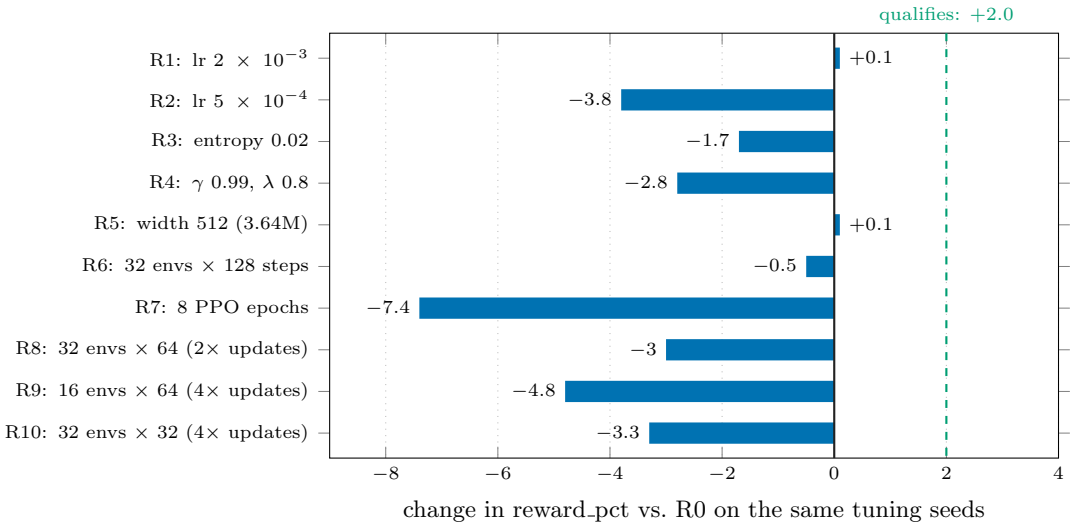


Figure 3: TASK-026, one-factor screen around GRU-TP at the EP-A budget (differences as listed in the report, section 8.1; R0 = 48.83 ± 0.45 reward on 4 seeds, 48.62 on the 2-seed subset used for most variants). **Reading:** nothing reaches the green line; every variant that gives the PPO learner more updates per environment step (R7–R10) is *worse*. By the pre-registered rule no final block was run.

3.2 Data-scaling diagnostic: the gap is sample efficiency

The same GRU-TP configuration was trained for more environment steps (labelled `diag-*`, learning-rate schedule scaled to the run length, 2 seeds beyond 1M).

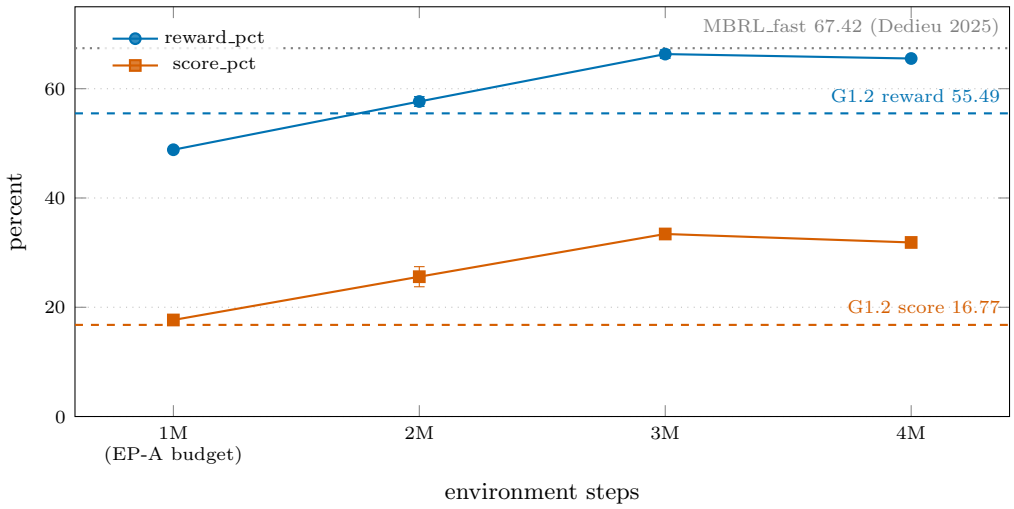


Figure 4: TASK-026 data-scaling diagnostic (mean ± SE; 1M: 4 seeds, others: 2 seeds, single configuration, CPU). **Reading:** the curve crosses both G1.2 lines between 1M and 2M steps (2M: reward $-2\,\mathrm{SE} = 55.87$) and plateaus near 66 reward from 3M. This is why the report concludes that the missing quantity is learning per environment step (about 1.7–2×), not model capacity. The shape, not the exact values, is the finding.

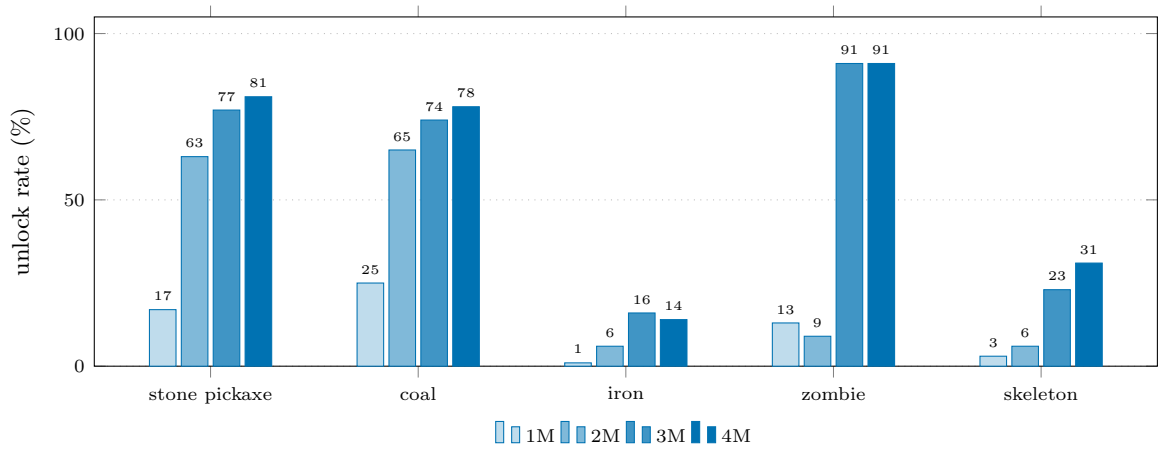


Figure 5: TASK-026, unlock rates of the deep achievement chain for GRU-TP at 1M, 2M, 3M, 4M steps (report section 8.2). **Reading:** the 1M policy is stuck at the start of the tech chain (iron ≈ 1%); the extra data is spent on stone tools, coal, then iron and the monsters. `eat_plant`, diamond and iron tools remain at 0% even at 4M (not drawn).

3.3 Ceiling argument (1M-step policies only)

Taking, for every achievement, the best unlock rate over the five 1M-step arms of TASK-025 gives a union of **52.63** reward (11.58 achievements per episode), still below 55.49 (12.21 per episode). The arms are complementary (GRU-TP raises `eat_cow` 45→75 %, Truck variants defeat zombies 47–59 % vs. 10–16 %), but no mix of 1M-step policies reaches the gate. This is a statement about the EP-A budget, not about what the model family can represent (Fig. 5).

3.4 Conclusions of TASK-026

- **Knob tuning is exhausted for G1.2** at 1M steps: ten single-knob variants, none above +0.1 reward.
- **The first-draft reading was corrected**: the pre-registration expected a missing capability; the diagnostic shows a sample-efficiency gap (README amendment 4).
- **No final block, no gate claim.** Candidate directions (none implemented, since each is an unconfirmed component): Dyna-style use of the TP head, an off-policy/high update-to-data learner, more TP-like auxiliary signals, image modality.

4 TASK-027: sample-efficiency ideas for G1.2

4.1 Design: three operator ideas and one addition

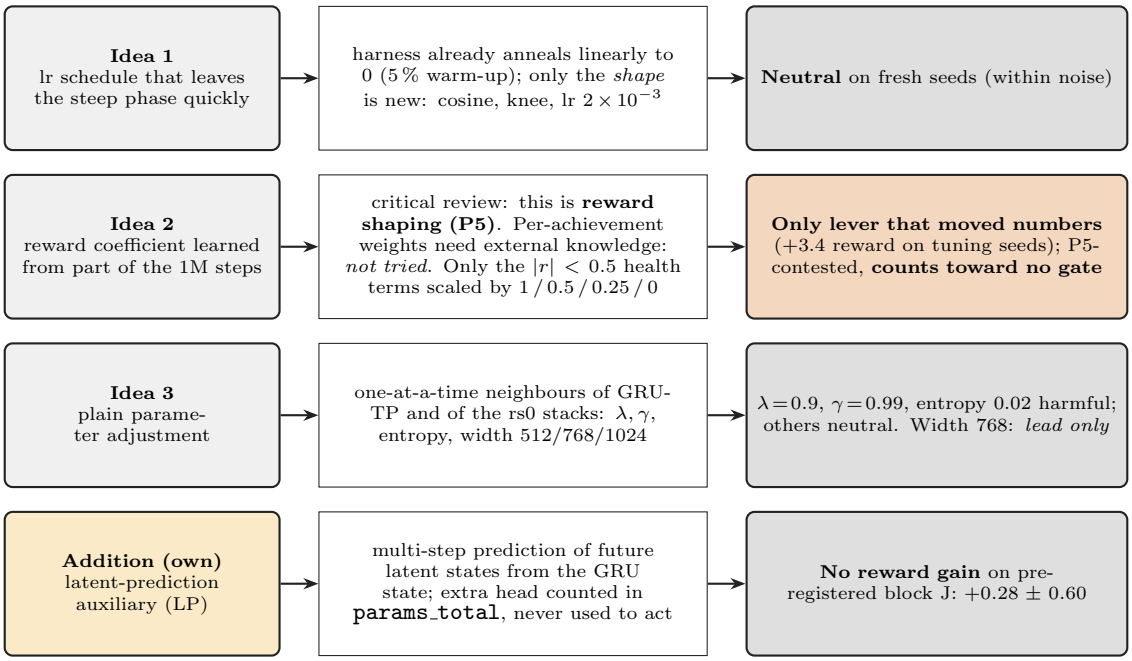


Figure 6: TASK-027 design and outcome, one row per idea. Idea 1 is the operator's wording; the coefficient in idea 2 was *not learned*: the effect is monotone and ends at 0, so a learned coefficient would have to converge to "ignore the health reward", and no principled criterion for that exists that does not already encode that the metric counts achievements.

4.2 Screening on tuning seeds: one round in detail

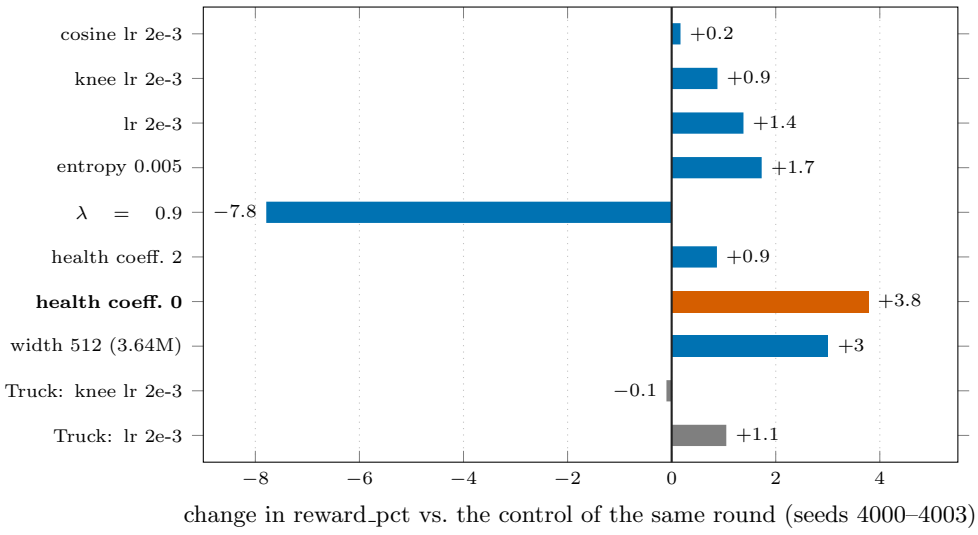


Figure 7: TASK-027 round R1 (4 seeds each; differences computed from Appendix A: GRU rows against `G4TP1` = 47.11, Truck rows against the Truck-plus-TP-head control `TP1h` = 45.94). **Reading:** only the health coefficient 0 and width 512 stand out. Please note the noise floor: the identical configuration and seeds gave 48.83 in TASK-026 (CPU) and 47.11 here (T4), a 1.7 difference, so differences below about 2 are not interpretable.

4.3 The reward-coefficient lever: monotone, but it is reward shaping

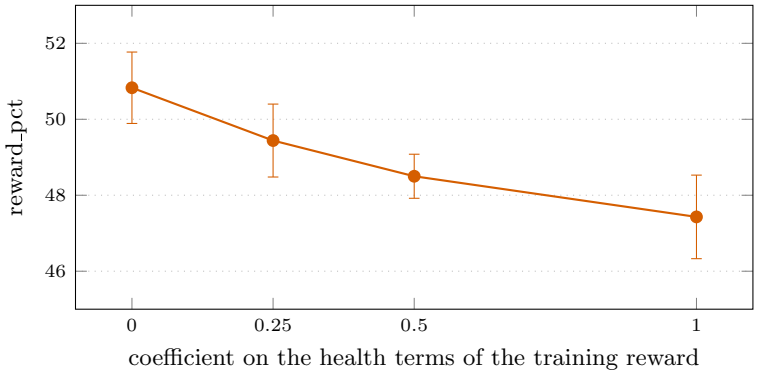


Figure 8: Dose-response of the health-term coefficient (rounds R2–R4, seeds 5000–5003, GRU-TP; mean ± SE). The coefficient 1 point is the unmodified control `G4TP1`. **Reading:** a clean monotone trend (47.4 → 50.8), reproduced on a second seed set (4000–4003: 47.11 → 50.89 at coefficient 1 → 0). Because the effect ends at 0, it is "ignore health reward": a protocol-level change, not a learning gain.

4.4 Screening gains shrink on fresh seeds (winner's curse)

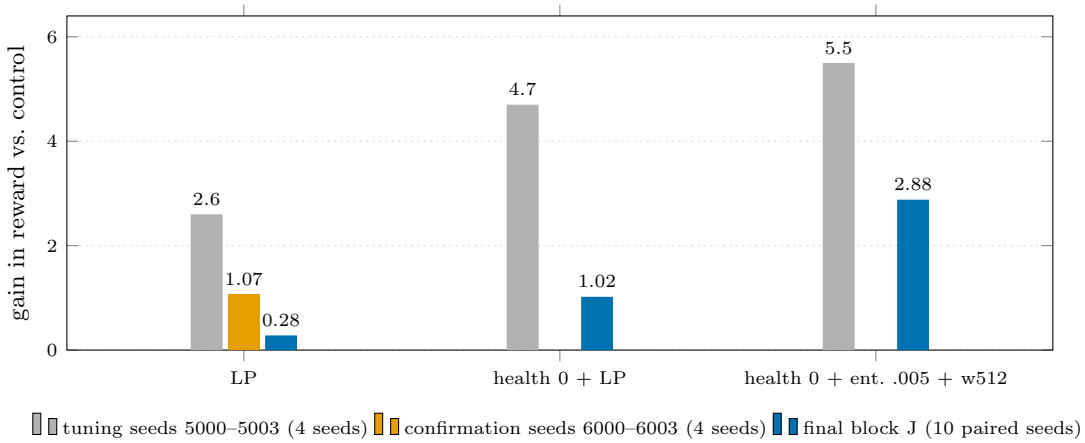


Figure 9: Gain over the GRU-TP control in the screening round versus the pre-registered final block. Tuning and confirmation gains are differences of 4-seed means (Appendix A); final gains are paired differences over 10 seeds (RESULTS section 5). **Reading:** every gain shrinks on fresh seeds, which is the expected behaviour of selecting the best of about 35 configurations. The LP gain falls from +2.60 to +0.28 ± 0.60 (not distinguishable from zero).

4.5 Final EP-A blocks

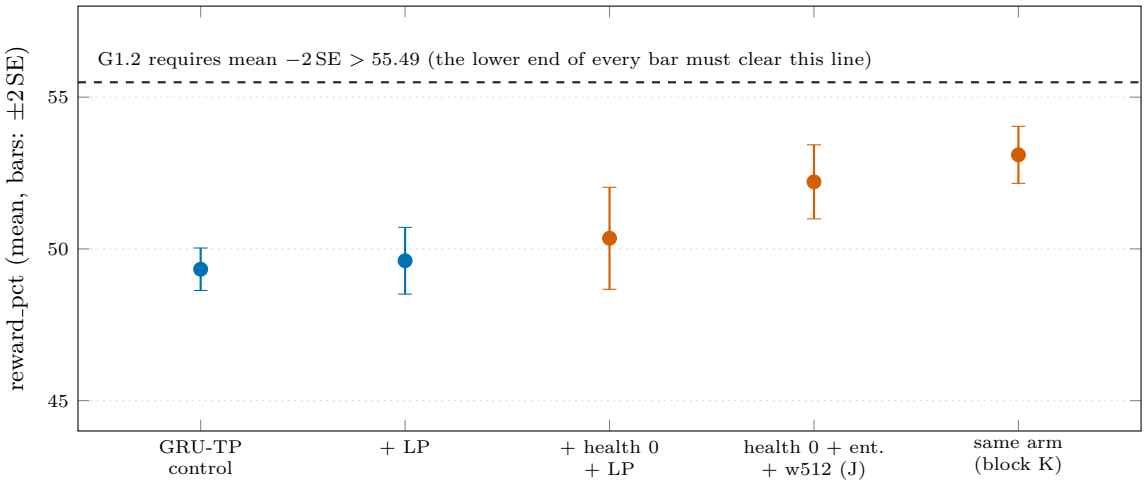


Figure 10: Final blocks, 10 seeds each: mean with a ±2 SE bar. Blue: P5-clean. Orange: P5-contested. **Reading:** no lower bar end comes within 3.3 reward of the dashed line. Blocks J and K for the last arm are independent seed sets (train 172–181 / test 554–563, and train 182–191 / test 564–573).

arm	protocol	params	reward (mean ± SE)	−2SE	score (mean ± SE)	−2SE	G1.2	gap[a]
`G4TP1` (control, J)	EP-A	1.25M	49.33 ± 0.35	48.62	19.01 ± 0.46	18.09	fail	6.87
`G_lp1` (J)	EP-A	1.52M	49.61 ± 0.55	48.52	19.88 ± 0.41	19.06	fail	6.97
`G_rs0_lp1` (J)	contested	1.52M	50.35 ± 0.84	48.66	20.28 ± 0.65	18.99	fail	6.83
`G_rs0_ent005_w512` (J)	contested	3.64M	52.21 ± 0.61	50.98	20.32 ± 0.78	18.76	fail	4.51
`G_rs0_ent005_w512` (K)	contested	3.64M	53.10 ± 0.47	52.16	21.86 ± 0.52	20.82	fail	3.33

[a] gap = 55.49 − (mean − 2 SE) on reward (arithmetic on the table values). Score clears 16.77 for every arm; reward is the only failing metric.

Paired differences to the same-seed control (block J, $n = 10$; seed s uses test seed $s + 382$ for every arm).

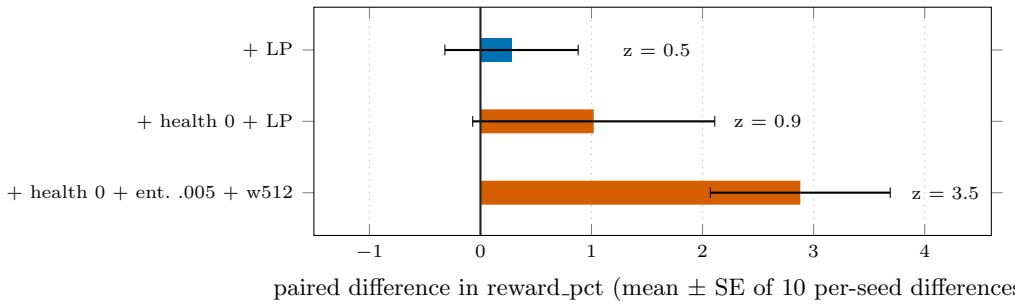


Figure 11: Paired reward differences in block J (RESULTS section 5). Score differences: +0.86±0.38 (z 2.3), +1.27±0.96 (z 1.3), +1.31 ± 1.07 (z 1.2). **Reading:** LP is indistinguishable from the control on reward; only the full contested stack shows a reward gain clearly above noise (+2.88 ± 0.81), and it is the one that violates P5.

4.6 Width round: a lead, not a result

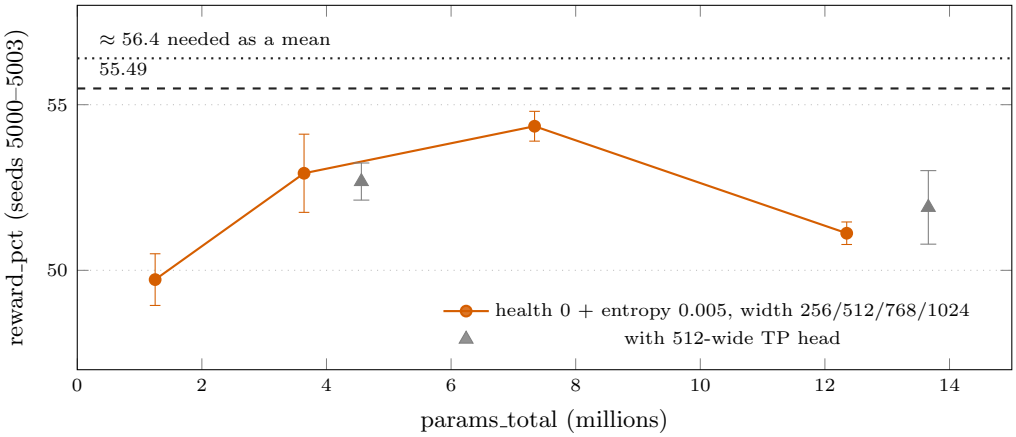


Figure 12: Round R7: width and TP-head size, all P5-contested, tuning seeds, $n = 4$ (mean ± SE). Width 768 reached 54.35 ± 0.45 reward / 23.75 ± 0.11 score, the highest tuning-seed number of the task, but the response is non-monotone (512: 52.93, 768: 54.35, 1024: 51.12), it is the best of four configurations screened on the same four seeds, and width 512 had already shrunk on fresh seeds. **No final block was run** (about 70 minutes of one kernel slot remained before the deadline), so width 768 is not gate evidence and, even if it held, would still need about +2 reward.

4.7 Conclusions of TASK-027

- **G1.2 is not reached.** Best clean arm: 49.61 (block J). Best contested arm: 52.21 (J) and 53.10 (K); 55.49 needed at mean -2 SE.
- **No P5-clean lever moved reward**: lr shape, lr 2×10^{-3}, λ, γ, entropy, width and the new LP auxiliary were neutral or harmful on fresh seeds. TASK-026 found the same for the plain knobs.
- **The one lever that moved numbers is reward shaping** (idea 2). It carries the labels `EP-A-P5-contested-reward-coefficient`, `gate_eligible=false` and separate `+rs0` arm labels in MLflow.
- **Distance remaining:** about 1.7× data efficiency (TASK-026 2M-step diagnostic 57.7); the best contested arm closes roughly +3 of the ≈ +9 that doubling the data gives (no efficiency factor is claimed from this).

5 What the two trials establish

Claim	Evidence	Strength and limits
Plain PPO knobs (lr, entropy, γ/λ, epochs, batch shape, width) do not close G1.2	TASK-026 (10 variants, 2–4 seeds); TASK-027 (about 35 configurations, 4 seeds)	Strong for "no large effect"; screens detect only effects at least about 2 reward
More PPO updates per env step hurt	TASK-026 R7–R10 (-3.0 to -7.4)	Consistent in four variants; one config
The gap is sample efficiency, not capacity	TASK-026 2M/3M/4M diagnostic (Fig. 4); width 512–1024 gives no clean gain	Moderate: 2 seeds, one configuration, CPU; shape not exact values
The LP auxiliary does not raise reward	block J, paired $n = 10$: $+0.28 \pm 0.60$	Strong (pre-registered, fresh seeds); score $+0.86 \pm 0.38$ is suggestive only
Zeroing the health reward raises reward by about 1–3	tuning seeds (+3.4), block J stack ($+2.88 \pm 0.81$)	Real but P5-contested; no gate value; smaller on fresh seeds
Width 768 might reach about 54	one 4-seed tuning set	**Lead only**: best of four, non-monotone, no final block
A *learned* reward coefficient	not built	Effect is monotone to 0, so no learning criterion was found that is not circular
Dyna-style imagination, off-policy high-UTD learner, image modality	not tested in either trial	Open; each needs a design decision

6 Caveats recorded in the task files

Caveat	Detail
Seed overlap	Tuning seeds 4000–4003 are shared by TASK-026 and TASK-027; they were never used for a gate.
Unequal tuning effort (P9)	The tuned arms have about 35 configurations × 4 seeds of screening; the control has none (TASK-025 configuration). This favours the tuned arms.
Winner's curse	Screening gains shrink on fresh seeds (Fig. 9); 2–4-seed means are optimistic by several points.
Not bit-reproducible	GPU runs differ run to run; across platforms the same configuration differed by 1.7 reward (Fig. 7).
Platforms differ	TASK-026 on a 4-core CPU (jax 0.12.0); TASK-027 on Kaggle T4 (jax 0.11.1). Within a block every arm ran on the same platform.
Modality	Our runs use the 1345-d symbolic observation; the paper thresholds use the 63×63×3 image, so calibration against them is approximate.
Diagnostics	The 2M–4M runs use 2 seeds and one configuration; they are `diag-*`, never a gate.
Open operator decisions	(a) Is any reward-coefficient arm admissible, i.e. amend P5? (b) Does GRU-TP's second-attempt G1.1 pass close G1.1?

7 Decisions we would kindly ask you to take

1. **P5.** Keep the rule (the reward-coefficient arms stay diagnostics) or amend it, in which case the coefficient must be declared and the baseline given equal effort in a fresh pre-registered block.

2. **G1.1 ruling.** Whether GRU-TP's pass on its second attempt closes G1.1 (the roadmap defines no retry rule; all blocks are reported, none dropped).

3. **Next direction for sample efficiency.** Candidates, none built: Dyna-style use of the TP head (supported by TP being the only auxiliary that helped, and by Dedieu et al. reaching 67.4 at 1M steps with a transformer world model), an off-policy/high update-to-data learner (PPO with more reuse hurt), further TP-like auxiliary signals (the contrastive EV auxiliary hurt, so the choice matters), image modality. Each requires a design decision and a GPU.

A Complete screening tables of TASK-027

Table 2: All screening results of TASK-027 (Kaggle T4, 4 seeds per row; copied mechanically from `RESULTS.md`, sections 3–4). Reward and score are mean ± SE; r+s is their sum.

round	config	params_total	reward	score	r+s	per-seed reward
R1 (seeds 4000-4003)	`G4TP1`	1,254,358	47.11 ± 0.79	17.39 ± 1.03	64.5	48.7 48.2 46.2 45.4
	`G_cos2`	1,254,358	47.28 ± 0.46	16.15 ± 0.38	63.4	47.2 46.5 48.6 46.8
	`G_ent005`	1,254,358	48.84 ± 0.61	18.81 ± 1.28	67.7	49.7 48.3 47.4 50.0
	`G_kn2a`	1,254,358	47.99 ± 0.94	18.99 ± 0.91	67.0	46.2 48.2 50.6 47.0
	`G_lam9`	1,254,358	39.32 ± 0.70	12.80 ± 0.50	52.1	38.0 40.9 38.3 40.1
	`G_lr2`	1,254,358	48.49 ± 1.07	17.75 ± 0.67	66.2	48.4 48.9 45.8 50.9
	`G_rs0`	1,254,358	50.89 ± 0.32	21.15 ± 0.91	72.0	51.7 50.8 50.1 50.9
	`G_rs2`	1,254,358	47.98 ± 0.20	17.03 ± 0.53	65.0	47.5 48.1 47.9 48.4
	`G_w512`	3,640,790	50.12 ± 0.66	19.55 ± 0.99	69.7	51.8 50.3 49.8 48.6
	`TP1h`	484,117	45.94 ± 0.49	16.52 ± 0.68	62.5	45.7 47.3 45.7 45.0
	`T_kn2a`	484,117	45.84 ± 2.38	16.48 ± 2.16	62.3	43.5 50.1 49.5 40.3
	`T_lr2`	484,117	46.99 ± 0.79	18.57 ± 0.35	65.6	48.5 48.2 45.7 45.5
R2 (5000-5003)	`G4TP1`	1,254,358	47.43 ± 1.10	17.39 ± 1.05	64.8	47.4 44.4 49.4 48.5
	`G_lp03`	1,521,622	48.39 ± 0.23	17.93 ± 0.93	66.3	48.8 47.7 48.4 48.7
	`G_lp1`	1,521,622	50.03 ± 1.48	20.56 ± 1.59	70.6	47.9 47.7 50.3 54.1
	`G_lp1k6`	1,521,622	48.63 ± 0.89	19.65 ± 0.62	68.3	46.5 50.5 49.7 47.9
	`G_lp3`	1,521,622	47.83 ± 0.29	19.44 ± 0.56	67.3	48.3 48.3 47.3 47.4
R3 (5000-5003)	`G_rs0`	1,254,358	50.83 ± 0.94	21.03 ± 1.03	71.9	51.5 51.7 48.0 52.1
	`G_rs0_ent02`	1,254,358	45.97 ± 1.64	17.78 ± 1.35	63.7	44.8 43.7 50.8 44.5

round	config	params_total	reward	score	r+s	per-seed reward
	G_rs0_g99	1,254,358	48.21 ± 0.70	18.27 ± 0.85	66.5	49.8 46.6 48.9 47.5
	G_rs0_kn2a	1,254,358	50.33 ± 0.51	20.13 ± 0.91	70.5	50.0 49.2 50.5 51.6
	G_rs0_lam9	1,254,358	40.30 ± 0.70	13.68 ± 0.45	54.0	39.3 39.3 40.3 42.3
	G_rs0_lr2	1,254,358	50.42 ± 0.61	20.42 ± 0.37	70.8	50.1 51.3 51.5 48.8
R4 (5000-5003)	G_rs025	1,254,358	49.44 ± 0.96	19.66 ± 1.02	69.1	52.2 49.0 47.9 48.6
	G_rs05	1,254,358	48.50 ± 0.58	18.82 ± 0.89	67.3	48.0 49.7 49.2 47.1
	G_rs0_ent005	1,254,358	49.72 ± 0.78	19.07 ± 1.42	68.8	47.4 50.8 50.1 50.5
	G_rs0_ent005_w512	3,640,790	52.93 ± 1.18	21.72 ± 1.59	74.6	49.6 55.2 53.6 53.3
	G_rs0_w512	3,640,790	51.40 ± 1.06	21.61 ± 0.98	73.0	53.3 52.7 50.9 48.6
	T_rs0	484,117	48.17 ± 2.10	18.97 ± 2.53	67.1	44.6 52.6 50.9 44.5
R5 (5000-5003)	G_lp1_ent005	1,521,622	49.65 ± 0.39	18.68 ± 0.34	68.3	50.8 49.0 49.3 49.5
	G_lp1_kn2a	1,521,622	47.21 ± 0.52	17.68 ± 1.16	64.9	46.4 47.4 46.4 48.6
	G_lp1_w512	4,170,710	47.91 ± 0.62	17.54 ± 1.27	65.5	47.9 46.9 47.2 49.6
	G_rs0_lp1	1,521,622	52.13 ± 0.51	22.63 ± 0.55	74.8	52.9 50.8 52.9 51.9
R5c (6000-6003)	G4TP1	1,254,358	48.54 ± 1.01	18.72 ± 1.13	67.3	50.1 50.5 47.1 46.5
	G_lp1	1,521,622	49.61 ± 1.01	19.01 ± 1.07	68.6	48.3 47.7 50.4 52.1
R6 (5000-5003)	G_lp1_w512_ent005	4,170,710	48.74 ± 0.61	16.33 ± 0.76	65.1	47.3 49.2 50.1 48.3
	G_rs0_lp1_ent005	1,521,622	52.32 ± 0.54	21.18 ± 0.81	73.5	53.5 52.1 50.9 52.8
	G_rs0_lp1_w512	4,170,710	51.60 ± 1.03	21.11 ± 0.57	72.7	51.5 48.7 53.4 52.8
	G_rs0_lp1_w512_ent005	4,170,710	52.34 ± 1.06	21.22 ± 0.81	73.6	52.1 53.3 54.5 49.5
R7 (seeds 5000-5003)	G_rs0_ent005_w1024	12,345,814	51.12 ± 0.34	20.28 ± 0.44	71.4	51.1 51.7 51.6 50.2
	G_rs0_ent005_w1024_tph512	13,656,022	51.90 ± 1.11	20.76 ± 1.13	72.7	54.4 52.4 49.0 51.9
	G_rs0_ent005_w512_tph512	4,557,782	52.68 ± 0.56	22.06 ± 1.39	74.7	51.7 51.8 54.0 53.3
	G_rs0_ent005_w768	7,337,942	54.35 ± 0.45	23.75 ± 0.11	78.1	54.8 53.0 54.6 54.9

B Per-seed values of the final block J

Table 3: Per-seed reward of the four arms in the final block J (train seeds 172–181; test seed = train seed +382). Copied from `RESULTS.md`, section 5.

arm	172	173	174	175	176	177	178	179	180	181
G4TP1	47.5	49.3	50.0	47.4	50.2	50.9	49.8	49.8	49.5	48.9
G_lp1	49.2	51.9	49.3	48.8	49.5	52.2	47.7	47.4	48.5	51.5
G_rs0_lp1	51.2	52.3	50.8	53.9	43.9	49.5	49.9	50.7	49.2	52.1
G_rs0_ent005_w512	53.7	53.6	53.1	52.8	49.1	53.1	48.4	52.6	51.6	54.0

C Sources and reproduction

- TASK-026: `docs/experiments/2026-10-19_phase1_g12_tuning/README.md` and `.../PHASE1_RESULTS_REPORT.md`; report command `scripts/run_t026_cpu_sweep.pyreport--stageR1` (also R3, D2M, D3M, D4M); ceiling script `scripts/adhoc/t026_gap_ceiling.py`.
- TASK-027: `docs/experiments/2026-10-22_sample_efficiency/README.md` (pre-registration with amendments 0–8) and `.../RESULTS.md`; final-block summary `pythonscripts/summarize_t026.py`.
- Raw result files: `output/phase1/g12_t026/` (TASK-026) and `output/phase1/push_t026/` (TASK-027; LFS, plain copies under each `results/`). MLflow: 206 runs ingested in this checkout, 0 UNREGISTERED.
- State entries: `docs/core/STATE.yaml`, tasks TASK-20261019-026 and TASK-20261022-027. Impact check: no LOCKED_CORE_THEORY symbol touched; new options default off.
- Differences in the figures that are not quoted directly (Figs. 7, 9) are subtractions of table values, as stated in each caption.